

Newton's Laws of Motion

Why things move the way they do: forces, mass, and the three laws that connect them.

By **Pratik Bhanuse**, PhD Physics candidate, Western University ·
pratikbhanuse.github.io

Kinematics tells us *how* things move. **Dynamics** tells us *why*. In 1687 Isaac Newton stated three laws that, together with a list of the forces that act, predict the motion of everything from a falling apple to a planet. They remain the working tools of every mechanical, civil, aerospace and biomedical engineer, accurate whenever speeds are well below the speed of light and objects are much larger than atoms. This chapter states the laws carefully, builds the most important skill in mechanics (drawing free-body diagrams), and applies it to connected bodies, pulleys, inclines and accelerating frames.

YOU WILL BE ABLE TO

- Identify the forces acting on a body and classify them (gravity, normal, tension, spring, friction).
- State Newton's three laws precisely, including the role of inertial frames.
- Draw correct free-body diagrams and apply $\sum \vec{F} = m\vec{a}$ component by component.
- Correctly identify third-law action–reaction pairs.
- Solve problems with connected bodies, pulleys and inclines, using constraint relations.
- Calculate apparent weight in accelerating lifts.
- Analyse motion in accelerating (non-inertial) frames using pseudo-forces.

THINK FIRST

A book rests on a table. The Earth pulls the book down with a force $\vec{W}$, and the table pushes it up with a force $\vec{N} = \vec{W}$. Is $\vec{N}$ the “reaction” to $\vec{W}$ in the sense of Newton’s third law?

No. Both forces act on the *same* object, the book, and third-law pairs always act on *different* objects. The reaction to the Earth’s gravitational pull on the book is the book’s gravitational pull on the **Earth**. The reaction to the table’s push on the book is the book’s push on the **table**. $\vec{N}$ equals $\vec{W}$ here only because the book is not accelerating (the second law). In a lift accelerating upward, $\vec{N} > \vec{W}$, while third-law pairs are always exactly equal.

§3.1

Forces and interactions

A **force** is a push or a pull, an interaction between two objects. It is a vector, measured in **newtons** ($1\text{ N} = 1\text{ kg m/s}^2$). Every force has an *agent*: something exerts it. If you cannot name the agent, the force does not exist.

At the deepest level there are only four fundamental interactions: gravitational, electromagnetic, and the strong and weak nuclear forces. In mechanics we meet just two of them, in many disguises:

FORCE	SYMBOL	ORIGIN	KEY FACTS
Weight (gravity)	$\vec{W} = m\vec{g}$	Gravitational pull of the Earth	Acts at the centre of gravity, vertically down, magnitude mg
Normal force	$\vec{N}$	Electromagnetic repulsion between surfaces in contact	Perpendicular to the surface; adjusts itself to prevent interpenetration; can only push
Tension	$\vec{T}$	Electromagnetic forces between	Along the string, pulling away from

FORCE	SYMBOL	ORIGIN	KEY FACTS
		molecules of a string, rope or cable	the body; can only pull
Spring force	$\vec{F}_s = -k \Delta \vec{x}$	Molecular forces in a deformed spring	Proportional to extension or compression (Hooke's law), opposite to the deformation
Friction	$\vec{f}$	Surface interactions	Parallel to the surface, opposing relative sliding (Chapter 4)

Contact forces (normal, tension, friction, spring) need touching. **Field forces** (gravity, electric, magnetic) act at a distance.

DEFINITION

Ideal string: massless and inextensible. The tension is the same everywhere along it, and two bodies joined by a taut ideal string have equal speeds and equal accelerations along the string. **Ideal pulley:** massless and frictionless. It changes the direction of a tension without changing its magnitude.

§3.2

Newton's first law and inertial frames

KEY IDEA

First law (law of inertia). A body stays at rest, or moves with constant velocity in a straight line, unless a net external force acts on it.

The tendency to keep the same velocity is called **inertia**, and **mass** is its quantitative measure. A loaded lorry is harder to start and harder to stop than a bicycle because it has more inertia.

The first law is not a special case of the second. Its real content is to **define the frames of reference** in which Newton's laws hold. An **inertial frame** is one in which a body with no net force acting on it moves with constant velocity. The ground is very nearly an inertial frame for most engineering purposes, and so is any frame moving at constant velocity relative to it. An accelerating car, a lift that is speeding up, and a rotating merry-go-round are **non-inertial**: in them, objects appear to accelerate with no force acting. We return to these in the last section.

IN PRACTICE

Seat belts and head restraints exist because of the first law. In a frontal crash the car stops in about 0.1 s, but an unrestrained occupant continues at the original speed until something (the steering wheel, the windscreen) exerts a force. In a rear-end collision the seat pushes the torso forward while the head, by inertia, lags behind. This is the mechanism of *whiplash*, which head restraints are designed to limit. Loose objects inside a car become projectiles for the same reason.

§3.3

Newton's second law

KEY IDEA

Second law. The net force on a body equals the rate of change of its momentum:

$$\sum \vec{F} = \frac{d\vec{p}}{dt}, \quad \vec{p} = m\vec{v}.$$

For a body of constant mass this becomes

$$\sum \vec{F} = m\vec{a}.$$

Points to understand about the second law:

- It is a **vector** equation. It means three independent equations: $\sum F_x = ma_x$, $\sum F_y = ma_y$ and $\sum F_z = ma_z$.

- $\sum \vec{F}$ is the **net** external force: the vector sum of *all* forces acting *on* the body. Forces the body exerts on other things do not appear.
- The acceleration is in the direction of the **net force**, not necessarily the direction of motion. A ball thrown upward moves up while its acceleration points down.
- The law is **instantaneous**: the acceleration at a given moment is set by the forces at that moment. When a force is removed, the acceleration vanishes at once, but the velocity does not.
- If $\sum \vec{F} = \vec{0}$, then $\vec{a} = \vec{0}$. The body is in **equilibrium**, either at rest or moving at constant velocity.

The form $\vec{F} = d\vec{p}/dt$ is more general, and we use it for rockets and other variable-mass systems in Chapter 6.

§3.4

Newton's third law

KEY IDEA

Third law. If body A exerts a force on body B, then B exerts a force on A that is equal in magnitude and opposite in direction:

$$\vec{F}_{A \text{ on } B} = -\vec{F}_{B \text{ on } A}.$$

Third-law pairs always:

1. act on **different** bodies, so they never cancel each other in a free-body diagram;
2. are of the **same type** (gravitational with gravitational, contact with contact);
3. act **simultaneously**. Neither is “first”.

COMMON MISTAKE

A common confusion: “If the horse pulls the cart and the cart pulls the horse back equally, how can they move?” The answer is that the two forces act on **different** bodies. Whether the cart accelerates depends only on the forces *on the cart*: the horse’s forward pull minus friction on its wheels. The horse moves forward because the **ground** pushes it forward, through friction on its hooves, harder than the cart pulls it back.

§3.5

Free-body diagrams and a problem-solving strategy

The free-body diagram (FBD) is the single most important tool in mechanics. It isolates one body and shows every force acting on it, and only those forces.

KEY IDEA

Strategy for Newton's-law problems.

1. **Choose the system.** Pick the body (or group of bodies) to analyse.
2. **Draw the FBD.** Draw the body as a dot or simple shape. For each force, name the agent and draw an arrow from the body. Check for gravity, then everything *touching* the body: surfaces (normal and friction), strings (tension), springs.
3. **Choose axes.** Align one axis with the acceleration if you know its direction. On an incline, use axes along and perpendicular to the surface.
4. **Write $\sum \mathbf{F} = m\mathbf{a}$ for each axis**, and for each body if there are several.
5. **Add constraint relations** linking the accelerations of connected bodies (strings, pulleys, surfaces).
6. **Solve algebraically**, then substitute numbers. Check units, signs and limiting cases.

COMMON MISTAKE

Never put “***ma***” on a free-body diagram as if it were a force. It is the *result* of the forces, not one of them. Never include forces the body exerts on other things. And “the force of motion” does not exist: a moving body needs no force to keep moving.

WORKED EXAMPLE 3.1

Pulling a box at an angle

A 4.0 kg box on a frictionless floor is pulled by a rope with a force of 20 N at 30° above the horizontal. Find the acceleration of the box and the normal force from the floor.

Solution

FBD. Four forces act on the box: its weight $m\mathbf{g} = 39.2$ N (down), the normal force $\mathbf{N}$ (up), the tension $\mathbf{T} = 20$ N at 30° above horizontal, and no friction. Take x horizontal (the

direction of motion) and y vertical.

x -direction:

$$T \cos 30^\circ = ma \Rightarrow a = \frac{20(0.866)}{4.0} = 4.33 \text{ m/s}^2.$$

y -direction. There is no vertical acceleration:

$$N + T \sin 30^\circ - mg = 0 \Rightarrow N = 39.2 - 20(0.5) = 29.2 \text{ N}.$$

Evaluate. The normal force is less than the weight, because the rope supports part of the box. “Normal force equals weight” is true only in special cases. If the pull were increased until $T \sin 30^\circ = mg$ (that is, $T = 78.4 \text{ N}$), the box would lift off and N would be zero.

§3.6

Weight, normal force and apparent weight

A bathroom scale does not measure your weight. It measures the **normal force** it exerts on you. When the scale accelerates, the two are different. That reading is your **apparent weight**.

For a person of mass m in a lift with upward acceleration a , Newton’s second law (up positive) gives

$$N - mg = ma \Rightarrow N = m(g + a).$$

- Lift accelerating upward, or decelerating while moving down: $a > 0$, so you feel heavier.
- Lift accelerating downward, or decelerating while moving up: $a < 0$, so you feel lighter.
- Free fall ($a = -g$): $N = 0$. This is “weightlessness”. Gravity is still acting, but nothing pushes back.

WORKED EXAMPLE 3.2

Apparent weight in a lift

A 60 kg student stands on a scale in a lift. What does the scale read (in newtons) when the lift (a) accelerates upward at 2.0 m/s^2 , (b) moves upward at constant speed, (c) accelerates downward at 2.0 m/s^2 , and (d) falls freely after the cable breaks?

Solution

The forces on the student are the weight mg (down) and the normal force N from the scale (up). With up positive, $N = m(g + a)$.

(a) $N = 60(9.8 + 2.0) = 708 \text{ N}$.

(b) $a = 0$, so $N = 60(9.8) = 588 \text{ N}$, equal to the true weight.

(c) $N = 60(9.8 - 2.0) = 468 \text{ N}$.

(d) $a = -9.8 \text{ m/s}^2$, so $N = 0$.

Evaluate. The scale reading depends only on the **acceleration**, not on the velocity. Moving up at constant speed feels exactly like standing still.

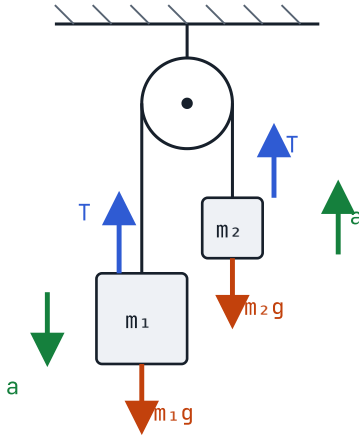
IN PRACTICE

Your inner ear is an accelerometer. The *otolith organs* of the vestibular system contain tiny calcium carbonate crystals resting on a gel layer above sensory hair cells. When your head accelerates, the crystals lag behind by inertia, bending the hair cells, and the brain reads the bending as acceleration. This is why you feel a lift start and stop but not its steady motion. It is also why astronauts in free fall, and patients with vestibular disorders, can lose their sense of “down”. Pilots are trained to trust their instruments over their inner ear, because a sustained, banked, accelerating turn can feel exactly like level flight.

§3.7

Strings, pulleys and connected bodies

When bodies are connected, apply $\sum \vec{F} = m\vec{a}$ to **each body separately**, then link the equations with **constraint relations**, which come from the fixed lengths of strings.



Atwood machine. The same tension T pulls up on both masses. The heavier mass accelerates down and the lighter one up, with the same magnitude of acceleration.

WORKED EXAMPLE 3.3

The Atwood machine

Masses $m_1 = 5.0$ kg and $m_2 = 3.0$ kg hang from an ideal string over an ideal pulley, as in the figure. Find the acceleration of the masses and the tension in the string.

Solution

Constraint. The string has fixed length, so if m_1 moves down with acceleration a , then m_2 moves up with the same a .

Second law for each mass, taking each one's direction of motion as positive:

$$m_1 : \quad m_1 g - T = m_1 a, \quad m_2 : \quad T - m_2 g = m_2 a.$$

Solve. Adding the equations eliminates T :

$$a = \frac{(m_1 - m_2)g}{m_1 + m_2} = \frac{2.0(9.8)}{8.0} = 2.45 \text{ m/s}^2.$$

Substituting back:

$$T = m_2(g + a) = \frac{2m_1 m_2 g}{m_1 + m_2} = \frac{2(5.0)(3.0)(9.8)}{8.0} = 36.8 \text{ N}.$$

Evaluate.

- **Limits.** If $m_1 = m_2$, then $a = 0$ (balanced). If $m_2 \rightarrow 0$, then $a \rightarrow g$ and $T \rightarrow 0$ (m_1 falls freely).

- **Size of T .** The tension lies between the two weights, $m_2g = 29.4$ N and $m_1g = 49$ N, as it must for m_1 to accelerate down and m_2 up.

Historically, Atwood used this device in 1784 to “dilute” gravity and measure g with slow, timeable motion.

WORKED EXAMPLE 3.4

Block on an incline pulled by a hanging mass

A 4.0 kg block rests on a frictionless incline at 30° . A string parallel to the incline runs from the block over a pulley at the top to a hanging 3.0 kg mass. Find the acceleration and the tension.

Solution

Which way does it move? Compare the driving forces along the string: the hanging weight $m_2g = 29.4$ N against the component of the block’s weight down the slope, $m_1g \sin 30^\circ = 19.6$ N. The hanging mass wins, so it descends and the block moves up the slope.

Equations. For the block, use axes along the incline (up-slope positive). For the hanging mass, take down as positive.

$$m_1 : T - m_1g \sin 30^\circ = m_1a, \quad m_2 : m_2g - T = m_2a.$$

Solve. Add the equations:

$$a = \frac{m_2g - m_1g \sin 30^\circ}{m_1 + m_2} = \frac{29.4 - 19.6}{7.0} = 1.4 \text{ m/s}^2.$$

$$T = m_2(g - a) = 3.0(9.8 - 1.4) = 25.2 \text{ N}.$$

Evaluate. For the block, the normal force is $N = m_1g \cos 30^\circ = 33.9$ N. It plays no part in the motion along the slope, because the incline is frictionless. If the slope were steeper, the block could win: setting $m_1 \sin \theta = m_2$ gives $\theta = 48.6^\circ$ for balance.

Bodies in contact

WORKED EXAMPLE 3.5

Pushing three blocks

Three blocks of masses 1.0 kg, 2.0 kg and 3.0 kg sit in a row on a frictionless floor, touching each other. A horizontal force of 12 N pushes on the 1.0 kg block. Find the acceleration and the contact forces between the blocks.

Solution

Whole system. All three blocks move together:

$$a = \frac{F}{m_1 + m_2 + m_3} = \frac{12}{6.0} = 2.0 \text{ m/s}^2.$$

Contact force between blocks 2 and 3. Draw the FBD of block 3 alone. Only F_{23} acts horizontally:

$$F_{23} = m_3 a = 3.0(2.0) = 6.0 \text{ N}.$$

Contact force between blocks 1 and 2. It must accelerate blocks 2 and 3 together:

$$F_{12} = (m_2 + m_3)a = 5.0(2.0) = 10 \text{ N}.$$

Check with block 1: $12 - 10 = 1.0(2.0)$. ✓

Evaluate. The force transmitted *decreases* along the chain, because each contact force has to accelerate only the blocks beyond it. Pushing from the other end, on the 3.0 kg block, would give contact forces of 10 N and 4.0 N instead. The arrangement matters.

Constraint relations with movable pulleys

When a pulley can move, the accelerations of the connected bodies are no longer equal. Find the relation from the **fixed length of the string**: write the string's length in terms of the positions of the bodies, then differentiate twice.

WORKED EXAMPLE 3.6

A movable pulley

Block A ($m_A = 2.0 \text{ kg}$) rests on a frictionless table. A string tied to A runs horizontally to a fixed pulley at the table's edge, down to a *movable* pulley, and back up to a hook on the underside of the table. Block B ($m_B = 2.0 \text{ kg}$) hangs from the movable pulley. Both pulleys and the string are ideal. Find the accelerations of A and B and the tension.

Solution

Constraint. Let x be A's distance from the fixed pulley and y the depth of the movable pulley below the table. Two string segments of length about y hang below the table, so the string length is

$$L = x + 2y + \text{constant}.$$

Differentiating twice gives $\ddot{x} + 2\ddot{y} = 0$. In magnitudes, when A moves towards the pulley with acceleration a_A , B descends with

$$a_B = \frac{a_A}{2}.$$

Forces. The same tension T acts along the whole string.

- Block A (horizontal): $T = m_A a_A$.
- Block B plus the movable pulley (down positive): two strands pull up, so $m_B g - 2T = m_B a_B$.

Solve. Substitute $T = m_A a_A$ and $a_B = a_A/2$:

$$m_B g - 2m_A a_A = \frac{m_B a_A}{2} \Rightarrow a_A = \frac{2m_B g}{4m_A + m_B} = \frac{2(2.0)(9.8)}{8.0 + 2.0} = 3.92 \text{ m/s}^2.$$

So $a_B = 1.96 \text{ m/s}^2$ and $T = m_A a_A = 7.84 \text{ N}$.

Check B: $m_B g - 2T = 19.6 - 15.7 = 3.92 \text{ N} = m_B a_B$. ✓

Evaluate. The movable pulley gives a 2:1 *mechanical advantage*: B is supported by $2T$, but moves only half as far as A. This trade of force for distance is the principle behind block-and-tackle hoists, and behind the pulley systems used for orthopaedic traction (Problem P3.8).

§3.8

Non-inertial frames and pseudo-forces

Newton's laws hold in inertial frames. Sometimes it is easier to work in an accelerating frame, for example the frame of a car, a lift or a centrifuge. In a frame with acceleration $\vec{a}_0$ relative to an inertial frame, Newton's second law still works *if* we add a **pseudo-force** (fictitious or inertial force) to every body:

$$\vec{F}_{\text{pseudo}} = -m \vec{a}_0.$$

Then $\sum \vec{F}_{\text{real}} + \vec{F}_{\text{pseudo}} = m \vec{a}_{\text{rel}}$, where $\vec{a}_{\text{rel}}$ is the acceleration measured in the accelerating frame.

A pseudo-force has no agent and no third-law partner. It is a bookkeeping device that accounts for the acceleration of the frame. The “push back into your seat” as a plane accelerates for take-off is the pseudo-force, felt in the plane’s frame.

WORKED EXAMPLE 3.7

A pendulum as an accelerometer

A 0.50 kg bob hangs from the roof of a car that accelerates forward at 3.0 m/s^2 along a level road. Find the angle the string makes with the vertical and the tension in the string. Solve it (a) in the ground frame and (b) in the car’s frame.

Solution

(a) **Ground frame (inertial).** The bob accelerates forward at $a = 3.0 \text{ m/s}^2$. Only two real forces act: the tension T , at angle θ to the vertical, and the weight mg .

$$\text{Horizontal: } T \sin \theta = ma, \quad \text{Vertical: } T \cos \theta = mg.$$

Divide the equations: $\tan \theta = a/g = 3.0/9.8 = 0.306$, so $\theta = 17.0^\circ$, with the bob hanging backwards. Square and add:

$$T = m\sqrt{g^2 + a^2} = 0.50\sqrt{9.8^2 + 3.0^2} = 5.12 \text{ N}.$$

(b) **Car frame (non-inertial).** The bob is at rest. Add a pseudo-force ma pointing backward. Then T , mg and ma balance, giving the same $\tan \theta = a/g$ and $T = m\sqrt{g^2 + a^2}$.

Evaluate. In the car, the bob behaves as if gravity were $g_{\text{eff}} = \sqrt{g^2 + a^2} = 10.2 \text{ m/s}^2$, tilted backwards. Measuring θ measures the car’s acceleration. That is exactly how a simple accelerometer works.

IN PRACTICE

Accelerometers and centrifuges. The MEMS accelerometer in your phone is a microscopic proof mass on silicon springs. When the phone accelerates, the springs must supply the force $m\vec{a}$, so they flex, and the change in capacitance between tiny comb-shaped electrodes is measured. Seen from the phone's frame, the mass is pushed by a pseudo-force $-m\vec{a}$. The same idea, with gravity included, is how the phone knows which way is down. A laboratory **centrifuge** spins samples so that, in the rotating frame, they feel a pseudo-force thousands of times larger than gravity. This separates blood cells from plasma in minutes rather than hours.

CHAPTER SUMMARY

- Forces are interactions with an agent. In mechanics: weight $m\vec{g}$, normal force, tension, spring force $-k\vec{x}$, friction.
- **First law:** no net force means constant velocity. It defines inertial frames.
- **Second law:** $\sum \vec{F} = d\vec{p}/dt = m\vec{a}$ (constant mass). It is a vector law, and the acceleration points along the *net* force.
- **Third law:** $\vec{F}_{AB} = -\vec{F}_{BA}$. The pair acts on different bodies and is never drawn on the same FBD.
- **Method:** isolate each body, draw its FBD, choose axes along the acceleration, write $\sum F = ma$ for each axis and body, and add constraint relations from string lengths.
- Apparent weight $N = m(g + a)$. Free fall means $N = 0$.
- Atwood machine: $a = (m_1 - m_2)g/(m_1 + m_2)$ and $T = 2m_1m_2g/(m_1 + m_2)$.
- In a frame with acceleration $\vec{a}_0$, add the pseudo-force $-m\vec{a}_0$ to every body.

Practice problems

Full step-by-step solutions are in the separate solutions PDF.

LEVEL A – CONCEPT CHECK

P3.1

A horse pulls a cart along a level road. By Newton's third law, the cart pulls back on the horse with an equal and opposite force. Explain clearly how the horse and cart can accelerate forward.

P3.2

A book rests on a table. List every force acting on the book, and give the third-law partner of each one, stating which body that partner acts on.

P3.3

An astronaut aboard the International Space Station, about 400 km up, is said to be “weightless”. Yet gravity there is still about 89% as strong as at the Earth's surface. Explain what “weightless” really means, using the idea of apparent weight.

P3.4

A light string passes over an ideal pulley and connects two unequal masses. Is the tension equal on both sides? Is it equal to the weight of either mass? Would your answers change if the pulley had significant mass and the string did not slip?

LEVEL B – STANDARD PROBLEMS

P3.5

A 10 kg crate is pulled up a frictionless 30° ramp by a rope parallel to the ramp, with an acceleration of 1.5 m/s^2 . Find the tension in the rope and the normal force on the crate.

P3.6

A lift car has mass 800 kg. Its steel cable can safely carry a maximum tension of 12 000 N. What is the greatest upward acceleration the lift can have? What is the greatest downward acceleration if the cable must never go slack?

P3.7

Two blocks on a frictionless table are joined by a light string: a 4.0 kg block behind and a 2.0 kg block in front. A horizontal force of 30 N pulls the front block. (a) Find the acceleration and the tension in the connecting string. (b) If the string breaks at 15 N, what is the largest force that can be applied to the front block? (c) What would the largest force be if it were applied to the 4.0 kg block instead?

P3.8

Orthopaedic traction. In a simple leg-traction set-up, a rope supports a 5.0 kg mass, then passes round a pulley attached to the patient's foot. The two rope segments leave the foot pulley symmetrically, at 25° above and 25° below the horizontal. Find the net force the rope exerts on the foot. How would the traction force change if the angle were reduced to 10° ?

P3.9

A 60 kg person stands on a bathroom scale in a lift. During the first few seconds of a trip, the scale reads 660 N. Near the end of the trip it reads 520 N. Find the acceleration (magnitude and direction) in each case. Is the lift going up or down? Can you tell?

P3.10

A small weight hangs from a thread inside a train carriage. While the train is accelerating, the thread hangs at a steady 8.0° from the vertical, displaced towards the rear of the train. Find the train's acceleration. What would the thread do if the train were braking?

P3.11

Two blocks sit on the two faces of a fixed, frictionless double incline, joined by a light string over a pulley at the apex: $m_1 = 5.0$ kg on a 30° face and $m_2 = 3.0$ kg on a 60° face. Find the acceleration of the system, its direction, and the tension.

P3.12

For the Atwood machine of Example 3.3, find the force exerted on the pulley by the rod that holds it to the ceiling. Explain why this force is less than the total weight of the two hanging masses.

P3.13

Crane rigging. A 1000 kg load hangs in equilibrium from two cables. One makes 30° with the vertical and the other 45° with the vertical, on opposite sides. Find the tension in each cable. Which cable must be rated for the larger load, and why?

LEVEL C – CHALLENGE PROBLEMS**P3.14**

A block of mass m rests on the frictionless inclined face (angle θ) of a wedge of mass M . The wedge sits on a frictionless floor. A horizontal force F is applied to the wedge so that the block stays at rest relative to the wedge. Find (a) the acceleration of the system, (b) the normal force between block and wedge, and (c) F . Solve the problem in both the ground frame and the wedge's frame.

P3.15

A mass $M = 4.0$ kg hangs from a movable pulley. One end of a string is fixed to the ceiling. The string passes down and under the movable pulley, up over a fixed pulley, and down to a hanging mass $m = 3.0$ kg. All pulleys and strings are ideal. Find the acceleration of each mass (with directions) and the tension in the string.

P3.16

A body of mass m , initially at rest, is acted on by a force that decays with time as $F = F_0 e^{-t/\tau}$. Find (a) its velocity as a function of time, (b) its limiting velocity, and (c) the distance it covers in the first time interval τ .

P3.17

A uniform rope of mass M and length L lies on a frictionless floor and is pulled at one end by a horizontal force F . (a) Find the tension in the rope at a distance x from the pulled end. (b) Now the same rope hangs vertically from a ceiling. Find the tension at a distance y below the ceiling. Why is the “same tension everywhere” rule for ideal strings violated in both cases?

P3.18

The Atwood machine of Example 3.3 ($m_1 = 5.0$ kg, $m_2 = 3.0$ kg) is mounted inside a lift that accelerates upward at 2.2 m/s^2 . Find the acceleration of each mass relative to the lift, and the string tension. Then find the acceleration of m_1 relative to the ground.

All worked examples and practice problems in these notes are original.

Textbooks for further reading and practice, with a chapter-by-chapter map, are listed in the References.

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